

SPL3 Project Proposal Form, 2026
Institute of Information Technology (IIT)
University of Dhaka

Name: Md. Abul Kalam Azad	
Roll: BSSE-1425	Phone: +880 1301921242
Project Title: BanglaFactGuard: A Multimodal Bangla Fact Checking System	
Motivation: The rapid spread of misinformation on social media and online news platforms has become a major challenge in Bangladesh. Fake or misleading news often spreads much faster than manual fact-checking organizations can verify it. In many cases, fabricated news is circulated using the names of trusted newspapers, making it difficult for users to determine authenticity. Since professional verification is time-consuming, misinformation can influence public opinion before accurate information becomes available. Therefore, there is a need for a scalable Bangla fact checking system that can provide fast preliminary analysis while incorporating expert validation for reliable final verification.	
Project Description: Overview: BanglaFactGuard is a web-based Bangla fact checking platform designed to analyze and verify textual and image-based news content. Users will be able to submit news titles, descriptions, images and source information through an interactive interface. The system will first generate an automated preliminary prediction by analyzing both textual and visual content through a multimodal verification pipeline. After the initial prediction, expert users will review the result and provide verification opinions through a credibility-based voting mechanism. The final verification result will be determined using weighted expert feedback. The system aims to combine automated analysis with human expertise to provide faster, transparent and more reliable Bangla fact verification. Additionally, the platform will support source-based verification by retrieving evidence from newspaper sources and comparing it with the submitted claim to determine authenticity. Problem Statement and Objectives: Current Bangla fact-verification processes are mostly manual and investigation-driven, making verification slow and difficult to scale. During this delay, misinformation spreads rapidly across social media platforms. Moreover, users often encounter fake news falsely attributed to trusted newspapers without an easy way to verify authenticity. Additionally, fully automated fake news detection systems are not always reliable, as existing AI-based detection methods still struggle to achieve highly accurate and trustworthy verification results.	

The objectives of this project are:

- Develop a web-based platform for Bangla fact verification using textual and image-based content.
- Implement an automated multimodal verification pipeline for generating preliminary fake/non-fake predictions.
- Design a credibility-based expert voting system for reliable final verification.
- Implement source-based evidence retrieval and similarity analysis by extracting related news from claimed newspaper sources and comparing it with the submitted news content to verify authenticity.
- Provide transparent verification results including AI predictions, expert opinions, credibility information and final verification results.

Proposed Solution:

- **Multimodal Fact Checking Using ML/DL Model**

Users will submit fact titles, textual content and images through the web platform. The system will analyze both textual and visual information simultaneously to generate a preliminary prediction indicating whether the fact is fake or non-fake.

- **Expert-Driven Validation**

The AI-generated prediction will be treated as an initial assessment rather than a final decision. Expert users will review the content and provide verification opinions to improve reliability and reduce incorrect predictions.

- **Credibility Scoring Mechanism**

A credibility-based voting system will be implemented for expert validation. Experts with a stronger history of accurate reviews will receive higher credibility scores, causing their opinions to have greater influence on the final verification result.

- **Source-Based Fact Checking**

Users will also be able to verify whether a claim was actually published by a specific newspaper or media outlet. The system will retrieve related evidence from the claimed source and compare it with the submitted news content.

- **Interactive Visualization**

The system will present AI predictions, expert opinions, credibility information and final verification results through an interactive and user-friendly interface with suitable statistics to ensure transparency and accessibility.

References:

- [1] Faria, F. T. J., Moin, M. B., Hasan, Z., Khandaker, M. A. A., Islam, N., Hasib, K. M., & Mridha, M. F. (2024). MultiBanFakeDetect: Integrating advanced fusion techniques for multimodal detection of Bangla fake news in under-resourced contexts. *International Journal of Information Management Data Insights*, 4(2). Elsevier.
- [2] Panchendrarajan, R., & Zubiaga, A. (2024). Claim detection for automated fact-checking: A survey on monolingual, multilingual and cross-lingual research. *Natural Language Processing Journal*, 7, 100066. Elsevier.

Supervisor's Name: Toukir Ahammed

Signature of the supervisor: _____

Date: _____

Before the Midterm Presentation 1:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Proposal Presentation Feedback (if any):

Midterm 1 Presentation Feedback (if any):

Midterm 2 Presentation Feedback (if any):

